Internet Appendix for
Do product market threats discipline corporate misconduct?
(Not to be published)

This Internet Appendix presents the results of additional analyses and robustness tests discussed in the main text. The tables are organized as follows:

Table IA1: Instrumental variables approach

Table IA2: Firm performance and the competition-misconduct relation

A. The instrumental variables approach

As a supplementary approach to alleviating potential endogeneity, we employ the IV method based on two-stage least square (2SLS) estimation. Following Xu (2012) and Li and Zhan (2019), we use two instruments, namely import tariff (*Tariff*) and foreign exchange rate (*Exchange rate*), to exploit exogenous variation in product market fluidity and use it to explain the misconduct variables.

To qualify as a valid instrument, the variable used needs to be strongly correlated with the instrumented regressor (i.e., the relevance condition) but uncorrelated with the error term (i.e., the exclusion condition). Both of our instruments satisfy the two conditions, respectively. First, import tariff is regarded as an important trade barrier for foreign rivals that works to reduce the pressure from foreign competition (Bernard et al., 2007; Tybout, 2003). On the other hand, the variation in import tariffs is arguably orthogonal to the characteristics of each firm's product space dynamics in the sense that tariffs do not reflect choices by individual firms. Second, exchange rate, expressed as the amount of foreign currency per U.S. dollar, is positively correlated with foreign competition because a higher exchange rate makes foreign goods cheaper in U.S. dollars and thereby encourages import (Xu, 2012). At the same time, it is plausible that the determination of exchange rates lays outside of the corporate sector. Given its floating nature, the dollar's exchange rates should be determined at the macroeconomic level and reflect primarily the aggregate demand and supply of factors such as the balances of payments between the U.S. and its trade partners, interest rates, and inflation, none of which is likely to be caused by individual firm characteristics.

Both instruments are defined at the three-digit SIC industry level. To construct the tariff variable, we start with the import tariff data compiled by Frésard and Valta (2016), which cover the years until 2005.[23] We then supplement this information with the import tariff data compiled by Schott (2008) to cover the period from 2006 to 2018.[24] Note that the tariff data are only available for manufacturing industries (SIC 200–399), we thus restrict our focus to these industries for the IV regressions. For each industry year, the ad valorem tariff rate is calculated as duties collected by the U.S. customs divided by the free-on-board value of imports. To construct the foreign exchange rate variable, for each industry, we compute the source-weighted average of exchange rates across all countries exporting to the U.S., where the weights are the share of each

[23] The data are available at http://www.valta.ch/.
[24] The data are available at https://sompks4.github.io/sub_data.html.

exporting country in total U.S. imports for that industry in 1996.[25] We choose 1996 as the base year since our sample begins in 1997. These weights are fixed over our sample period because, as indicated by Xu (2012), most industries have relatively stable country distribution in the import share. We obtain the exchange rate data from the International Financial Statistics of the International Monetary Fund (IMF).

It is noteworthy that our regressions include both industry and year fixed effects, thereby removing any cross-industry differences and any common time series changes in the level of misconduct. As a result, the identification relies completely on industry-specific changes in tariffs and foreign exchange rates.

The IV regressions are conducted in two stages. In the first stage, we regress *Fluidity* on tariffs, exchange rates, firm controls, and fixed effects. The results are reported in column 1 of Table IA1. As expected, both tariffs and exchange rates are significant in predicting product market fluidity. The negative coefficient estimate on *Tariff* suggests that higher tariff rates reduce competition from foreign rivals, and the positive coefficient estimate on *Exchange rate* is consistent with the view that higher exchange rates make foreign goods cheaper and in turn encourage foreign competition. In addition, the *F*-statistic of 133.14 rejects the null hypothesis that the instruments are jointly zero. The fact that our instruments are significantly related to market competition suggests the relevance condition is satisfied.

In the second stage, we replace *Fluidity* with the predicted value of *Fluidity* from the first-stage regression and estimate its relation to the misconduct variables in columns 2 and 3 of Table IA1. The coefficient estiamtes on the fluidity variable in both specifications are negative and significant, confirming the negative relation between market competition and misconduct. Moreover, with more than one instrumental variable, an overidentification test can assess whether the instruments are correlated with the regression errors, which would undermine instrument validity. In each regression, the Hansen *J*-statistic is insignificant, indicating that the overidentification test fails to reject the hypothesis that the instruments are valid. This, along with other robustness checks, enhances confidence in the credibility of the empirical strategy.

Comparing the results obtained from the OLS regressions in Table 3 with those obtained from the above 2SLS regressions, it is interesting to observe that the magnitudes of the 2SLS

[25] Following Xu (2012), we drop countries that account for less than 2% of total U.S. imports in a given industry.

coefficient estimates are larger than those of the OLS estimates (even though the coefficient estimates from both approaches are negative and statistically significant), suggesting that OLS regressions bias the coefficient estimates upward due to endogeneity in product market competition. In other words, the omitted variables simultaneously make competition lower and misconduct less likely. One example of such omitted variables is CEOs' long-term orientation. Long-term thinking CEOs could self-select into non-competitive industries to the extent that the lower short-term pressure gives them more leeway in achieving long-term objectives. At the same time, they could also be less inclined to engage in misconduct, resulting in a spurious positive relation between competition and misconduct. In turn, this positive relation due to the omitted variable biases the coefficient estimates of *Fluidity* upward. Once we use the instruments to clean up the spurious correlation, the endogeneity of market competition is alleviated, and the coefficient estimates become more negative.

Table IA1. Instrumental variables approach

This table presents the two-stage least squares (2SLS) regression results using the instrumental variables approach. *Violator* is an indicator variable set to one if a firm has at least one violation in a year, and zero otherwise. *Penalties* is the natural logarithm of one plus the total value of penalties for violations in 2010 dollars. The instrumental variables for competition include the import tariff rate and the foreign exchange rate at the three-digit SIC level. Because these IVs are available for manufacturing firms (SIC 200-399) only, the regressions in this table are conducted on manufacturing firms. Column 1 reports the first-stage regression results, and columns 2 and 3 present the second-stage regression results. All other variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles, and dollar values are expressed in 2010 dollars. Industry and year fixed effects are included. Statistical significance is based on the heteroscedasticity-robust firm-clustered standard errors reported in parentheses. ∗∗∗, ∗∗, and ∗ indicate significance at the 1%, 5%, and 10% levels, respectively.

	First stage	Second stage	
Dependent variable	*Fluidity*	*Violator*	*Penalties*
	(1)	(2)	(3)
Fluidity		-0.125***	-0.153**
		(-5.09)	(-2.36)
Tariff	-2.872***		
	(-17.69)		
Exchange rate	0.015***		
	(3.62)		
Size	0.135***	0.370***	0.727***
	(5.32)	(22.35)	(17.24)
Leverage	-1.074***	-0.262**	-0.666***
	(-6.39)	(-2.17)	(-3.06)
M/B	0.140***	-0.019	0.008
	(8.29)	(-1.16)	(0.57)
PPE/TA	-2.984***	0.793***	0.868**
	(-10.13)	(3.87)	(2.21)
ROA	-1.839***	0.023	-1.006***
	(-17.90)	(0.16)	(-7.05)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	31,875	31,875	31,875
F-statistic	133.14		
P-value of *J*-statistic		0.14	0.19

Table IA2. Firm performance and the competition-misconduct relation

In this table, we split the sample into quartiles based on ROA in Panel A and M/B in Panel B, and re-estimate the baseline regression for each quartile. The dependent variables are as follows. *Violator* is an indicator variable set to one if a firm has at least one violation in a year, and zero otherwise. *Penalties* is the natural logarithm of one plus the total value of penalties for violations in 2010 dollars. The main explanatory variable of interest is *Fluidity*, a measure of product market threats constructed by Hoberg et al. (2014). All other variables are defined in Appendix A. Continuous variables are winsorized at the 1st and 99th percentiles, and dollar values are expressed in 2010 dollars. All specifications include industry and year fixed effects. Industries are defined based on the two-digit Standard Industrial Classification (SIC) codes. Statistical significance is based on the heteroscedasticity-robust firm-clustered standard errors reported in parentheses. ∗∗∗, ∗∗, and ∗ indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Quartiles based on ROA

Dependent variable	*Violator*	*Penalties*	*Violator*	*Penalties*	*Violator*	*Penalties*	*Violator*	*Penalties*
	1st Quartile (Highest ROA)		2nd Quartile		3rd Quartile		4th Quartile (Lowest ROA)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fluidity	-0.027**	-0.025***	-0.053***	-0.097***	-0.044***	-0.075***	-0.059***	-0.102***
	(-2.49)	(-4.04)	(-6.19)	(-6.54)	(-4.64)	(-3.14)	(-6.73)	(-5.07)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,825	16,048	16,028	16,042	16,032	16,045	16,027	16,038
Pseudo/Adjusted R^2	0.342	0.134	0.305	0.202	0.263	0.234	0.284	0.238

Panel B. Quartiles based on M/B

Dependent variable	*Violator*	*Penalties*	*Violator*	*Penalties*	*Violator*	*Penalties*	*Violator*	*Penalties*
	1st Quartile (Highest M/B)		2nd Quartile		3rd Quartile		4th Quartile (Lowest M/B)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fluidity	-0.033***	-0.053***	-0.049***	-0.095***	-0.061***	-0.100***	-0.071***	-0.098***
	(-3.31)	(-3.37)	(-5.40)	(-4.87)	(-6.94)	(-5.86)	(-6.71)	(-6.86)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,000	16,048	16,022	16,042	16,019	16,045	15,928	16,038
Pseudo/Adjusted R^2	0.303	0.199	0.294	0.237	0.307	0.244	0.392	0.249